Houssam Assaad

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Research Interests

Autonomous navigation of ground, aerial and legged robots; state estimation and sensor fusion (VI-SLAM, LiDAR-inertial); learning-based decision making and control (reinforcement learning); multivariable control; inspection and field robotics.

Education

American University of Beirut (AUB), Lebanon **Aug 2023 – May 2027 (expected)**
B.E. Mechanical Engineering, Minor in Economics; USAID HES Scholar *Beirut, Lebanon*

- **Relevant coursework:** System Analysis and Design (graduate, MECH 653), Control Systems, Instrumentation, Dynamics, Fluid Dynamics, Thermodynamics, Mechanical Design, Product Design and Development, Manufacturing Processes

Research Experience

King Fahd University of Petroleum & Minerals (KFUPM) **Jun 2026 – Sep 2026**
Research Intern, Control & Instrumentation Eng., supervisor Dr. Ramy Rashad *Dhahran, Saudi Arabia*

- Integrated a complete GNSS-denied sensing and estimation stack for a UAV: NVIDIA Jetson Orin Nano, Pixhawk 6X Pro (PX4) and Intel RealSense D435i, running ROS 2 Humble over a uXRCE-DDS bridge.
- Calibrated the IMU intrinsically, cutting gravity-magnitude error from 0.069 to 0.002 m/s^2, and characterized its noise with a 3-hour Allan-variance study (gyroscope bias instability 1.7–4.4 °/h).
- Benchmarked stereo-inertial SLAM (RTAB-Map) against OptiTrack motion capture over three trajectories: **0.100 m ATE RMSE**, path length within 6.8%; established a reproducible benchmarking protocol.
- First-author paper under review at the *Journal of Undergraduate Research International* (KFUPM).

American University of Beirut – Vision & Robotics Lab **Feb 2026 – Aug 2026**
*Undergraduate Research Assistant, supervisor Dr. Naseem Daher – **completed*** *Beirut, Lebanon*

- Formulated severity-aware pothole avoidance for autonomous ground vehicles as a Markov decision process; built discrete and continuous environments sharing one damage model.
- Developed tabular Q-learning, steering-only DQN and joint steering-and-speed DQN agents, with curriculum training and held-out test maps.
- Designed an exact-optimal evaluation baseline by backward dynamic programming over the finite acyclic state graph, defining a map-invariant *Normalized Damage Score* (NDS).
- DQN agent reached **NDS 88.8% ± 1.9%** on trained hazard densities and 51.4% on unseen densities (10 seeds), vs. 13.9% for the tabular baseline; the speed-extended agent modulates speed with hazard severity (NDS 83.6%).

Publications

H. Assaad, R. Rashad, "Benchmarking onboard visual–inertial SLAM against motion-capture ground truth," *Journal of Undergraduate Research International* (KFUPM), under review, 2026.

Experience

Middle East Airlines (MASCO) **Nov 2025 – Dec 2025**
Aircraft Maintenance Intern *Beirut, Lebanon*

- Trained on aircraft systems (avionics, airframe and powerplant) and toured maintenance facilities and workshops at Beirut Airport.

Projects

Autonomous Inspection Quadruped (Final Year Project) | *Unitree Go2, ROS 2, Nav2* **Sep 2026 – May 2027**

- Lead the inspection pipeline in a four-person team (advisor Dr. Hussein Hussein) building a quadruped that walks through a building, checks each element against the building's BIM model and reports what is missing or misplaced.
- Developing BIM-referenced LiDAR-inertial localization (adapting open-source LIO-BIM), a probabilistic semantic voxel map initialized from the BIM, and cost-aware navigation on Nav2.

- Integrating vision-language-model verification of inspected elements and coordinating interfaces with the teammates' camera arm and next-best-view planner. *(Ongoing.)*

Self-Balancing Robot: SISO to MIMO Control (MECH 436 & 653) | *Simulink, ESP32* **Nov 2025 – Present**

- **SISO (MECH 436):** designed and built a two-wheeled self-balancing robot, modeled it as an inverted pendulum and tuned a PID controller against sensor noise and actuator saturation; stabilized the hardware under applied load.
- **MIMO (MECH 653, graduate, ongoing):** extending to independent wheel torques that control pitch, forward velocity and heading; deriving a 6-state Lagrangian model and decoupling it into pitch and yaw subsystems.
- Designing state feedback with observers on an ESP32, analysing observability across sensor sets (IMU, wheel encoders), and benchmarking disturbance rejection and path following against OptiTrack.

Product Design & Development (MECH 525) | *Customer needs, benchmarking, prototyping* **Fall 2026**

- Five-person team taking a product from opportunity to a working prototype and a market pitch; screened 15 candidate opportunities with interviews, surveys, competitor and patent benchmarking and a weighted decision matrix.
- Converging on a low-cost 3D-scanning workflow for residual-limb capture and prosthetic socket design, targeting the access gap in prosthetic care (WHO: only 1 in 10 people who need prosthetic or orthotic devices have access). *(Ongoing.)*

NXP Cup Autonomous Racing Vehicle | *ROS 2, Gazebo, C++* **Feb 2025 – May 2025**

- Assembled the competition kit (sensors, actuators, control board, power) and built a modular ROS 2 stack with separate perception, localization, planning and actuation nodes.
- Optimized a minimum-time racing line under vehicle-dynamics constraints, validated in Gazebo and on track; joined inter-university training at Beirut Digital District.

Autonomous Irrigation System | *LabVIEW, myDAQ, Arduino* **May 2025**

- Led the design of a smart irrigation system using solar-radiation and soil-moisture sensing, with closed-loop control, real-time monitoring and manual/digital override in LabVIEW.

Microfluidic Mixing Analysis | *ANSYS, CNC, 3D Printing* **Jan 2025 – May 2025**

- Modeled two-inlet microfluidic mixing analytically (PDEs) and in CFD (ANSYS); fabricated CNC-machined micro-channels and validated them with dyed-fluid flow tests.

Go-Kart Design & Build | *SolidWorks FEA, Machining* **Jun 2025 – Aug 2025**

- Designed and validated a drive shaft for stress and fatigue in SolidWorks FEA; machined, assembled and tested a full-scale go-kart through iterative build–test cycles.

Leadership & Activities

AUB Robotics Club (leadership role, 2023 – present): organized competitive events (Build It 8.0); led an Arduino workshop series for 30+ students; mentor junior members on hardware, circuits and code.
Community service: 64+ hours per semester as a USAID scholar. **Model United Nations; Varsity sports:** represented AUB in Jiu-Jitsu and Taekwondo.

Skills

Robotics & autonomy: ROS 2 (Humble/Jazzy), Nav2, Gazebo, RTAB-Map / VI-SLAM, LiDAR-inertial SLAM, PX4/Pixhawk, NVIDIA Jetson, OptiTrack
Control & estimation: state-space modeling, PID and state feedback, observers, IMU calibration and Allan variance
AI & RL: Q-learning, DQN, dynamic programming, PyTorch, Gymnasium
Programming & tools: Python, C/C++, MATLAB/Simulink, Embedded C (ESP32, Arduino), Ubuntu, Git, LabVIEW
Design & manufacturing: SolidWorks (FEA), ANSYS, CNC machining, 3D printing
Languages: Arabic (native), English (fluent; IELTS Academic 8.0), French (elementary)